



Component Life Cycle

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2. Mounting phase
3. Updating phase
4. Unmounting phase
5. useEffect hook

- In mounting phase component is created and inserted into the DOM.
- The methods called in this phase are
- `constructor()`:
 - It is called when the component is initiated
 - It is the natural place to set up the initial state and other initial values.
- `static getDerivedStateFromProps()`
 - This is a static function that does not have access to 'this'.
 - It is always called just before the render method
- `render()`
 - It is the only required method
 - It returns JSX that is displayed in the UI.

- `componentDidMount()`
 - `componentDidMount()` is called as soon as the component is mounted and ready.
 - Good place to initiate API calls, if you need to load data from a remote endpoint.
 - It allows to update the state of the component if needed which will cause one more rendering.

- A component is updated whenever there is a change in the component's state or props.
- React has 5 built in methods which get executed one after another in the same sequence.
- `getDerivedStateFromProps()`
 - first method that is similar like mounting phase
- `shouldComponentUpdate()`
 - This lifecycle method should be very rarely used, and it exists only for certain performance optimizations.
 - This method returns a boolean value where default is true
- `render()`
 - It is called to re-render the HTML to the DOM, with the new changes.

- `getSnapshotBeforeUpdate()`
 - It is called right before the DOM is updated.
 - This method should also be used rarely or not used at all.
- `componentDidUpdate()`
 - This method is updating the DOM in response to prop or state changes.
 - `setState()` method can be called only if there is modification otherwise it may lead to infinite recursion

- The next phase in the lifecycle is when a component is removed from the DOM
- Only one built-in method that gets called when a component is unmounted: `componentWillUnmount()`
- Cleanup actions like clearing page refreshes can be executed in this lifecycle method.
- Call to `setState()` should not be done during this lifecycle method.

- `useEffect()` hook accepts 2 arguments:
`useEffect(callback[, dependencies]);`
- Callback is a function that contains the side-effect logic.
- Dependencies array
 - if empty, callback gets executed only once equivalent to `componentDidMount`
 - if omitted, callback gets executed for every update equivalent to `componentDidUpdate`
 - if specified with comma separated state members, callback gets executed for update of only those state members
- Callback method can return a function which gets executed as `componentWillUnmount()` life cycle method before component gets unmounted